Homework #7 (30 points) **Due:** Mar. 18, 2026, Wednesday

Purpose: This homework is to get familiar with SQL and ORACLE SQL-Plus Environment.

1. Create the Mail-order database (for the schema details and data, see the file '*MailOrder-DB-tables-info.pdf*') and populate it with the data in the Oracle SQL PLUS environment. Note: The column ***level*** in PARTS table needs named differently such as ***plevel*** since ***level*** is a keyword in Oracle SQL. **No blank lines inside CREATE TABLE commands** (Oracle SQLPlus is sensitive to blank lines in CREATE TABLE, gives a syntax error).

2. Your SQL commands should be placed in a file and run from the file

   - one file for creating tables, another file for inserting tuples, and another one for queries.

   - Use *Notepad++* editor (or *Microsoft .NET* editor) to create and edit your files.

   - Use 'sql' suffix for file names (i.e., queries.sql)

   Sample create table commands are included in the attachment.

3. In creating the tables, make sure to specify primary key and referential integrity constraints. You need to create tables referenced by foreign keys first.

4. Execute queries 2.1 − 2.5 and 2.7 − 2.10. Write down queries in a separate file and execute them from the file (queries are included in the attachment).

Use 'commit' command after every create table command, and after every insert group per table.
Use 'spool' command to get a log of screen (similar to 'script' of Unix).

Partial solution:
`K:\Courses\2026-Spring\ITC341-ugur1a\_Global-Materials_\hw7-partial\`
Also, in Blackboard. Make copies of the files and edit each file and add the rest of your solution.

How to put the screenlog into the file 'hw7-out.txt'

First, you need to create '**itc341**' directory in your U drive. Then, in U:\itc341, create a folder named **hw7**. Place all your files in **U:\itc341\hw7** (copy all files in **hw7-partial** folder into **hw7** folder). You also need to put your name, ITC 341 assignment number on top of every file as comments, like

```
-- Jim Goodwill
-- ITC 341     Homework 7     Mar. 18, 2026
```

set linesize 120 -- needed for displaying larger line output
set echo on -- **this is a MUST** (for including your SQL command and its result in the output file)
set pagesize 120 -- for a better display

You need to test your SQL files individually first. After a successful test, below is the necessary SQL-Plus commands you need to use for the final run:

SQL> spool U:\itc341\hw7\hw7-out.txt

-- run commands from your files, one by one, in the following order

SQL> start U:\itc341\hw7\cr-mailorder.sql -- assuming create table commands in U drive, -- '**cr-mailorder.sql**' file
SQL> start U:\itc341\hw7\ins-mailorder.sql -- assuming insert table commands in U drive, -- **ins-mailorder.sql**' file
SQL> start U:\itc341\hw7\q-mailorder.sql -- assuming queries in U drive, '**q-mailorder.sql**' file

SQL> spool off

The run results would be in the file 'U:\itc341\hw7\hw7-out.txt'.

What to hand in:

1. First make sure the results are correct (i.e., testing needs to be completed first).
2. **Submit the entire hw7 folder.** Copy the entire hw7 folder into

   K:\Courses\2026-Spring\ITC341-ugur1a\22482905\yourCMUglobalId

   K:\Courses\2026-Spring\ITC341-ugur1a\22482905\yourCMUglobalId folder is the submit drive. Each student has a folder under his/her CMU global id (**yourCMUglobalId** folder) with appropriate permissions.

3. Zip your hw7 folder as **yourCMUglobalId–hw7.zip** and submit to the Blackboard as well.

Sample create, insert commands (for the MAILORDER database)

```
-- cr-mailorder.sql
-- creates the mailorder DB-schema
-- ITC 341 partial hw7 (Homework #7)

-- keep these three commands at the top of every sql file
set echo on
set linesize 150
set pagesize 120

drop table ZIPCODES cascade constraints;
commit;

create table ZIPCODES
(
      zip char(5),
      city varchar(20),
      constraint ZIPPK        primary key(zip)
);

drop table EMPLOYEES cascade constraints;
commit;

create table EMPLOYEES
(
      eno number,
      ename varchar(20),
      zip char(5),
      hdate date,
      constraint EMPSPK    primary key(eno),
      constraint EMPSZIPFK
         foreign key(zip) references ZIPCODES(zip)
           ON DELETE SET NULL
);

-- ADD the rest of create table commands here
```

-- **ins-mailorder.sql**
-- inserts tuples into the mailorder DB
-- ITC 341 partial hw7

```
-- keep these three commands at the top of every sql file
set echo on
set linesize 150
set pagesize 120

delete from ZIPCODES;
commit;
insert into ZIPCODES values ('67226', 'Wichita');
insert into ZIPCODES values ('60606', 'Fort Dodge');
insert into ZIPCODES values ('50302', 'Kansas City');
insert into ZIPCODES values ('54444', 'Colombia');
insert into ZIPCODES values ('66002', 'Liberal');
insert into ZIPCODES values ('61111', 'Fort Hays');


delete from EMPLOYEES;
commit;
insert into EMPLOYEES values (1000, 'Jones', '67226', '12-DEC-95');
insert into EMPLOYEES values (1001, 'Smith', '60606', '01-JAN-92');
insert into EMPLOYEES values (1002, 'Brown', '50302', '01-SEP-94');

-- ADD the rest of insert into commands here
```

-- **q-mailorder.sql**
-- queries
-- ITC 341 partial hw7

```
-- keep these three commands at the top of every sql file
set echo on
set linesize 150
set pagesize 120

select * from employees;
select * from zipcodes;
select count(*) from employees;
select count(*) as DEPT_COUNT from zipcodes;

-- you may comment out the above queries for your homework

-- your queries goes here for hw#6

-- query 2.2: get all the rows in employees table
select * from employees;

-- You need to ADD the rest of queries here
```